

Valid responses per seed (of 20)

Learning rule / inference regime

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BPTT	0.0, 0.0, 5.0, 10.0, 15.0, 15.0, 20.0, 20.0
RFLO	0.0, 5.0, 5.0, 5.0, 5.0, 10.0, 15.0, 15.0, 20.0
PC (20 steps)	0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0
PC (100 steps)	0.0, 0.0, 0.0, 5.0, 5.0, 10.0, 20.0, 20.0

A scatter plot showing the relationship between 'Best training loss (log scale)' on the x-axis and 'Valid responses per seed' on the y-axis. The x-axis is logarithmic, ranging from 10^{-4} to 10^{-2} . The y-axis is linear, ranging from 0.0 to 20.0. The legend indicates four data series: BPTT (blue circles), RFLO (orange circles), PC (20 steps) (red circles), and PC (100 steps) (green circles). BPTT and RFLO generally achieve higher valid responses (up to 20) at higher training losses (around $10^{-2.5}$ to 10^{-2}). PC (20 steps) and PC (100 steps) generally achieve lower valid responses (mostly 0 or 5) across the range of training losses, with PC (100 steps) showing a slight increase in response at lower training losses (around $10^{-3.5}$ to 10^{-3}).

Best training loss (log scale)	Valid responses per seed (BPTT)	Valid responses per seed (RFLO)	Valid responses per seed (PC (20 steps))	Valid responses per seed (PC (100 steps))
$10^{-3.8}$			0.0	20.0
$10^{-3.7}$			0.0	5.0
$10^{-3.6}$			0.0	5.0
$10^{-3.5}$				20.0
$10^{-3.2}$				10.0
$10^{-2.8}$	20.0			
$10^{-2.7}$	20.0			
$10^{-2.5}$			0.0	
$10^{-2.4}$			0.0	
$10^{-2.3}$	15.0		0.0	
$10^{-2.2}$			0.0	
$10^{-2.1}$	15.0	15.0	0.0	
$10^{-2.0}$	10.0	10.0	0.0	
$10^{-2.0}$		5.0	0.0	
$10^{-1.9}$	5.0	5.0	0.0	
$10^{-1.8}$	0.0		0.0	

Valid Ready-Set-Go responses per seed across learning rules. (A) BPTT and RFLO produce valid responses reliably; PC fails at 20 inference steps and partially recovers at 100. (B) Best training loss (log scale) does not predict valid behavior. Points are individual seeds; bars in A are per-variant means.